GapSmooth: exact stochastic vibration inference across a sensor outage

Abstract

Missing displacement measurements create uncertainty about both displacement and velocity in vibrating systems. This reproducible software study asks how much future measurements reduce uncertainty inside a known outage under a linear Gaussian model. An exactly sampled stochastic oscillator is combined with Joseph-form Kalman filtering and Rauch–Tung–Striebel smoothing. Independent batch conditioning and covariance quadrature verify different parts of the calculation. In the recorded synthetic experiment, smoothing reduces midpoint displacement variance by 25.44%, while retaining nonzero uncertainty. An ensemble of 256 independent trajectories checks fixed-time calibration. Euler refinement separates time-discretization error from discrete inference error. Results establish consistency with the stated model, not physical validity.

1 Question, model and assumptions

A sensor outage does not freeze a structure: unobserved forcing continues to perturb its state. Future measurements can constrain an earlier trajectory, but cannot remove all uncertainty. The research question is therefore a conditional one: for specified dynamics, noise and sampling, how much smaller is the offline posterior variance than the causal filtering variance? The work connects state-space and Kalman-estimation concepts identified in the local DTU advanced time-series lectures with a mechanical vibration mode. No private course material, solutions or data are redistributed.

Let $z = (x, v)^T$, with displacement x in metres and velocity v in metres per second. A mass-normalized damped mode obeys the Itô SDE

$$dz = Az\,dt + L\,dW, \qquad A = \begin{bmatrix} 0 & 1 \\ -\omega^2 & -2\zeta\omega \end{bmatrix}, \quad D = LL^T = \begin{bmatrix} 0 & 0 \\ 0 & q \end{bmatrix}. \tag{1}$$

Here $\omega > 0$, $\zeta > 0$ and $q \geq 0$; q has units $\mathrm{m^2 s^{-3}}$. Observations are $y_k = Hz_k + \epsilon_k$, $H = [1, 0]$, with independent $\epsilon_k \sim \mathcal{N}(0, R)$ and $R > 0$. The initial Gaussian state, process noise and sensor noise are mutually independent. Parameters and the missing-data schedule are known. The model excludes nonlinear restoring forces, colored forcing and parameter uncertainty. The white-noise forcing is an idealization, not a measured load.

2 Numerical method and implementation

Linear SDE transition moments are available without time-stepping truncation [1]. For interval h,

$$F(h) = e^{Ah}, \qquad Q(h) = \int_0^h e^{As} D e^{A^T s}\,ds, \qquad z_{k+1} \mid z_k \sim \mathcal{N}(Fz_k, Q). \tag{2}$$

The implementation exponentiates $h\left[\begin{smallmatrix} A & D \\ 0 & -A^T \end{smallmatrix}\right]$; its upper blocks are F, B, and $Q = BF^T$. Before exponentiation the interval is scaled until $h\|A\|_\infty \leq 1$, then reconstructed by repeated composition $F \leftarrow F^2$, $Q \leftarrow FQF^T + Q$ using the previous F, Q. This avoids large unstable auxiliary blocks for long intervals. Matrices are symmetrized to remove antisymmetric roundoff; eigenvalues are not clipped.

For each time, predict $m^- = Fm$, $P^- = FPF^T + Q$. If a measurement exists, $S = HP^-H^T + R$, $K = P^-H^T/S$, and

$$m = m^- + K(y - Hm^-), \qquad P = (I - KH)P^-(I - KH)^T + KRK^T. \tag{3}$$

Missing observations skip the update. The observed-data negative log likelihood sums $[\log(2\pi S) + (y - Hm^-)^2/S]/2$ over available observations only. The backward smoother uses $G_k = P_k F_k^T (P_{k+1}^-)^{-1}$, $m_k^s = m_k + G_k(m_{k+1}^s - m_{k+1}^-)$ and $P_k^s = P_k + G_k(P_{k+1}^s - P_{k+1}^-)G_k^T$ [2]. Linear solves compute gains; explicit matrix inversion is avoided.

3 Reproducible experiment

The original Python package separates dynamics, recursive inference, a dense reference and experiment generation. NumPy and SciPy supply compiled small-matrix operations; a separate C++ binding is unnecessary for this two-state experiment. The recursive algorithm has linear work and storage in record length for fixed state dimension. The dense reference constructs the full joint state Gaussian and conditions all observed displacements in one solve; it is limited to 256 times because of quadratic storage and cubic solve cost.

The experiment uses $\omega = 2\pi$ rad/s, $\zeta = 0.08$, $q = 0.6$, $R = 0.02^2$ m^2 and $t_k = k/16$ for $k = 0, \ldots, 128$. Measurements at $3 \le t_k \le 5$ s are removed (33 samples). The zero-mean initial state has the analytical stationary covariance

$$P_\infty = \operatorname{diag}\left(\frac{q}{4\zeta\omega^3}, \frac{q}{4\zeta\omega}\right). \tag{4}$$

The prior is at the first supplied time, before its observation. Synthetic truth is generated from exact transition moments; seed 2718 gives Fig. 1. Filtering is causal; smoothing conditions on the entire record through 8 s. The sampling interval differs from any hidden integration step: there is none.

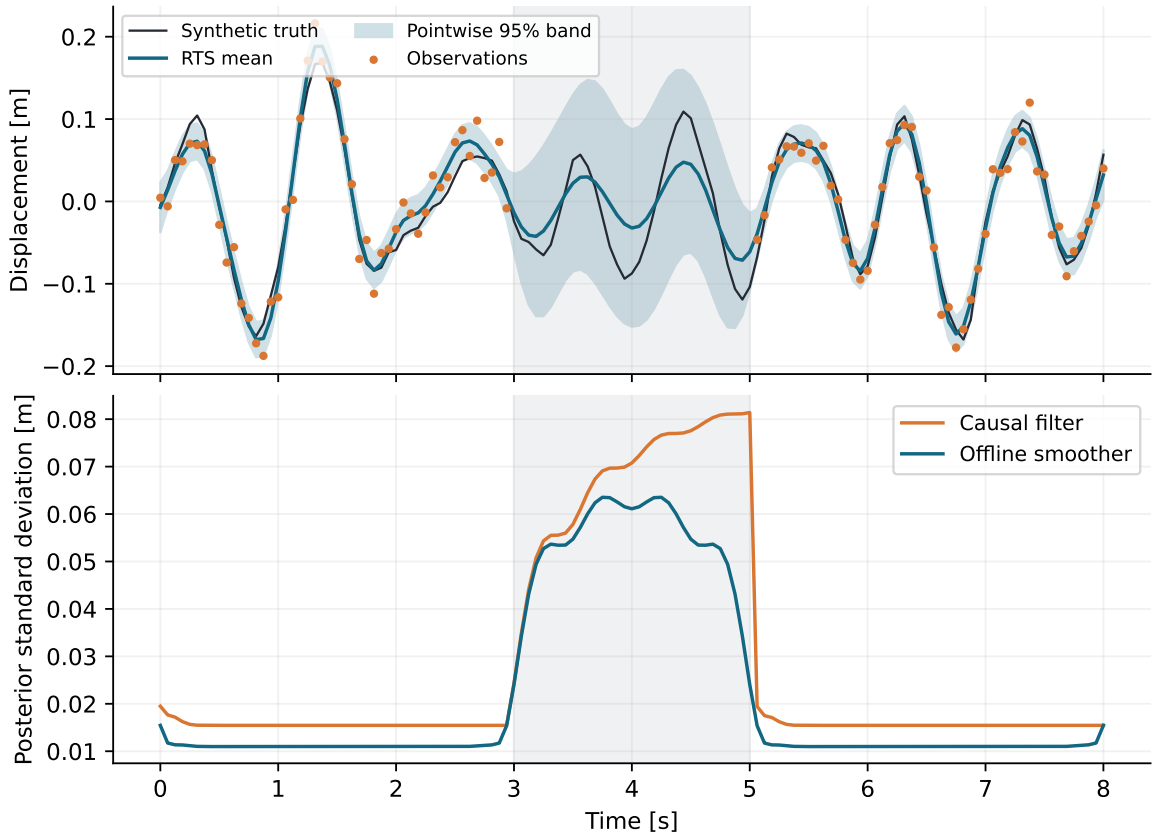


Figure 1: Original synthetic trajectory and uncertainty. Shading marks the missing-measurement interval. The blue band is a pointwise 95% smoothed displacement interval, not a simultaneous trajectory band. Future observations reduce uncertainty near both sides of the gap; the middle remains uncertain.

At $t = 4$ s the filtered displacement standard deviation is 0.07077 m and the smoothed value is 0.06111 m, a variance reduction of 25.44%. Because this is a linear model with fixed noise, both posterior covariance sequences depend on the observation schedule but not the observed values. This reduction is not a claim that smoothing improves every realized squared error.

An ensemble of 256 independent trajectories uses seeds 10000–10255 and the same prior, dynamics and outage. Only $t = 4$ s is used for calibration statistics: temporal samples are not counted as independent trials. For state error e, the normalized estimation error squared is $e^T P^{-1} e$ (NEES). Its expected value is two; the ensemble mean has a 95% model-consistency interval $[1.762, 2.252]$ from $\chi^2_{512}/256$. Observed means are 1.949 (filter) and 1.912 (smoother). Displacement coverage of pointwise 95% intervals is 96.88% and 97.66%, respectively. These finite-sample diagnostics are correlated between methods and do not prove calibration outside the generating model.

4 Verification and separation of errors

Discrete inference. On a nine-time irregular grid with four missing samples, the maximum differences from dense Gaussian conditioning are 5.551×10^{-16} for state means and 1.249×10^{-16} for marginal covariances; the likelihood difference is 8.882×10^{-16}. Prefix-conditioned batch estimates also verify causal filtering. This checks the recursion but shares transition moments, so it cannot alone verify Eq. (2).

Continuum moments. Independent adaptive quadrature of the covariance integral at $h = 0.3$ s differs by at most 1.388×10^{-17}; the stationary identity $FP_\infty F^T + Q = P_\infty$ differs by 2.220×10^{-16}. Additional tests cover semigroup composition, $h = 0$, tiny and long intervals, zero process intensity, absent observations, input errors and covariance reduction. Exact sampled moments have floating-point error, but no time-step truncation for Eq. (1). No continuum sample-path accuracy is claimed.

For contrast, forward Euler uses $F_E = I + hA$ and $Q_E = hD$. Composing these over one second yields Table 1. Errors use Frobenius norms with coordinates scaled by 1 m, 1 m/s and 1 s; their values depend on this scaling. This compares deterministic moments, not different random trajectories.

Table 1: Recorded Euler errors against exact one-second transition moments.

Steps	h [s]	$\lVert F_E^{\text{composed}} - F(1) \rVert_F$	$\lVert Q_E^{\text{composed}} - Q(1) \rVert_F$
16	0.062500	2.57861	0.40797
32	0.031250	0.73195	0.14898
64	0.015625	0.33249	0.06210
128	0.007812	0.16029	0.02834
256	0.003906	0.07867	0.01354

The recorded moment errors decrease as the Euler step is refined. Coarse steps can be unstable; this is not an unconditional stability or asymptotic-order claim.

5 Discussion, limitations and conclusion

The measured variance reduction quantifies the benefit of future information for one known model and schedule. It does not eliminate uncertainty caused by unobserved forcing. Discrete posterior agreement, continuum moment verification, and model-based coverage address distinct questions; none validates a physical structure. Experimental data, uncertain parameters, nonlinear mechanics and model mismatch remain outside scope. Covariance-form filtering can lose accuracy in extreme conditioning; a square-root implementation would require separate verification. The work claims no new filtering theory or peer review.

Reproduce with `pip install -e ".[dev]"`, then `gapsmooth -output reproduced`. Recorded metrics, CSV trace and figures are versioned. Tests, lint, package builds and sample reproduction are automated. Run `python paper/build.py -engine pdflatex` to generate data-bound numbers, compile, check references/layout warnings and enforce the five-page limit. A hash manifest links the PDF to code and numerical inputs; page rendering supports a separate human-style visual inspection.

References

[1] S. Särkkä and A. Solin. *Applied Stochastic Differential Equations*. Cambridge University Press, 2019. Chapters 6 and 10. Author-hosted text.

[2] S. Särkkä and L. Svensson. *Bayesian Filtering and Smoothing*. 2nd ed., Cambridge University Press, 2023. Author-hosted text.